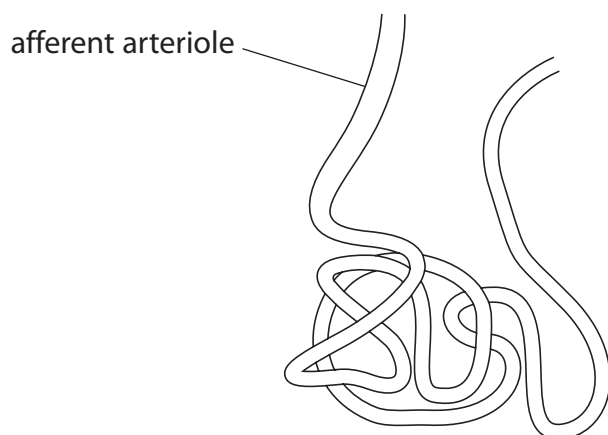


2 The kidney is an organ that is involved in the removal of waste metabolites from the blood.

(a) The drawing shows the blood supply inside a Bowman's capsule (renal capsule).



(i) Complete the diagram to show the Bowman's capsule and label **two** structures in this diagram.

(3)

(ii) Which substances are normally filtered from the blood in the Bowman's capsule?

(1)

- ☐ **A** amylose and glucose
- ☐ **B** glucose and sodium chloride
- ☐ **C** potassium ions and haemoglobin
- ☐ **D** urea and prothrombin

(iii) Which row shows the route taken by a water molecule through a nephron in the kidney?

(1)

Route through a nephron →				
☐ A	collecting duct	proximal tubule	distal tubule	loop of Henle
☐ B	loop of Henle	proximal tubule	collecting duct	distal tubule
☐ C	proximal tubule	distal tubule	loop of Henle	collecting duct
☐ D	proximal tubule	loop of Henle	distal tubule	collecting duct

(b) Describe how glucose reabsorption takes place in the nephron.

(3)

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(c) Urea is produced by the metabolic breakdown of some organic molecules.

Which statement describes the production of urea?

(1)

- ☐ A in the kidneys from excess amino acids
- ☐ B in the kidneys from glycogen
- ☐ C in the liver from excess amino acids
- ☐ D in the liver from glycogen

(Total for Question 2 = 9 marks)